

Bharath Raju

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PROFESSIONAL SUMMARY

Results-driven Data Analytics Engineering professional with 5+ years of experience in software development, DevOps, and data analytics across technology and financial services sectors. Expertise in Java, Python, AWS, and CI/CD pipelines, with a proven track record of delivering automated solutions that improved system reliability by 90% and reduced deployment time by 30%. Seeking to leverage advanced technical skills and process optimization expertise to drive data-informed decision making and operational excellence for growth-focused organizations.

EDUCATION

George Mason University

Master of Science in Data Analytics Engineering

Fairfax, VA

May 2025

Atria Institute of Technology

Bachelor of Technology in Computer Science Engineering

Bangalore, India

September 2020

PROFESSIONAL EXPERIENCE

George Mason University

Full Stack Developer

Fairfax, VA

November 2023 – May 2025

- Maintained and updated DFOR and TCOM departmental websites using Django templates and static asset versioning, ensuring consistent functionality with no significant outages, serving 1,000-2,000 monthly users including faculty and students, through systematic content management and technical updates.
- Engineered UI/UX enhancements using Django, HTML, CSS, and JavaScript including homepage layout simplification and breadcrumb navigation achieving noticeable page load improvements, receiving positive faculty feedback on content findability, through responsive design optimization and static asset optimization.
- Optimized SQL query performance in SQLite for dynamic tables including faculty publications and course listings reducing admin view load times by 15-20%, enhancing data accessibility for low-traffic administrative pages, through database indexing and Django ORM optimization.
- Maintained secure REST APIs for internal faculty data exposure including publications and contact information implementing input validation and endpoint security, ensuring secure data access for local modules pulling from static JSON and SQLite, through API security protocols and production code hygiene.

NTT DATA (Client: US Bank)

Software Engineer II

Bangalore, India

April 2022 – June 2023

- Spearheaded DevOps transformation initiative enhancing system reliability and automating 90% of deployment tasks using Jenkins and shell scripts within CI/CD pipelines, reducing deployment errors by 60% and improving release frequency by 200% from bi-weekly to 2-3 times weekly, through automated pipeline architecture coordinated across QA, DevOps, and cloud infrastructure teams.
- Led infrastructure upgrades and migrations of legacy batch processes and microservices to AWS EC2 and EKS contributing to 25-30% reduction in deployment time through Terraform and Docker automation, achieving 99.95%+ system availability through cloud optimization strategies, via container optimization and AWS resource management including S3 lifecycle policies and resource tagging.
- Administered Linux-based cloud environment monitoring Red Hat Linux VMs and AWS CloudWatch dashboards for system health resolving 90% of runtime issues escalated to engineering, maintaining zero P1 incidents during final 3 months and improving mean time to resolution by 35-40%, through proactive monitoring protocols using ServiceNow and Nexthink with automated alerts and root cause analysis playbooks.
- Developed scalable microservices using Java and Spring Boot for credit approval systems advancing system functionality and reducing average response time by 20-25% through Redis caching and query optimization, supporting traffic growth from 2K to 18K daily API calls during high-volume events, through microservices architecture with GitLab CI deployment.

NTT DATA

Software Engineer I

Bangalore, India

October 2020 – April 2022

- Designed enterprise applications using Java and Spring Boot for employee access management and HR system integration boosting processing speed and decreasing execution time by 10-15% through efficient code refactoring and architectural enhancements, supporting peak usage of 3,000 concurrent users during internal rollouts, through performance optimization techniques with Postman API testing and ServiceNow incident tracking.
- Developed API-based services for internal NTT tools integrating ticketing and asset management systems optimizing data flow and enhancing API performance by 20-25% through reduced payload and streamlined user access workflows, enabling traffic volume to double during onboarding season (January-March), through RESTful architecture implementation with DynaTrace and Grafana monitoring.

- Deployed containerized applications using Docker containers and Kubernetes (EKS) for internal services used by multiple departments ensuring high availability and reducing service downtime through automated scaling and YAML configuration with Helm charts, achieving 99.9%+ uptime across monitored environments and supporting elastic scaling for 150 containers, through container orchestration with AWS CloudWatch logging and SCCM integration.
- Implemented containerization and automated deployment using Docker, Bitbucket Pipelines, and Nexus for artifact storage minimizing environment inconsistencies by 90% and accelerating deployment cycles with scripted automation, reducing deployment time from 3-4 hours to 20 minutes per service and eliminating configuration drift through standardized container base images, through DevOps automation with infrastructure-as-code templates in Terraform for VM provisioning.

PROJECTS

- EcoQueryAI – Climate Data Access via LLM** **January 2025 - May 2025**
- Engineered modular, asynchronous MCP server using Python, FastMCP, and SQLite enabling real-time access to 15+ climate datasets via validated SQL queries, integrating structured JSON outputs with ClimateGPT LLM boosting AI response accuracy by 30% and reducing hallucination, through improved prompt engineering and Docker deployment on AWS EC2.
- AirScanAI - Big Data AQI Prediction Platform** **February 2024 - April 2024**
- Analyzed 29,531 air quality records from multiple Indian cities using Apache Spark and Databricks achieving R^2 of 0.89 and RMSE of 52.6 for AQI forecasting, demonstrating 75% pollution reduction during 2020 lockdowns in Delhi and Mumbai through time-series analysis potentially informing policymakers, through scalable big data processing.
- LoanWise - ML Loan Approval Platform** **January 2024 - February 2024**
- Designed end-to-end ML application for loan approval prediction simulation using publicly available dataset achieving 95.1% accuracy with Random Forest and 94.9% with XGBoost models, projecting millisecond processing time vs. traditional manual review and improving accuracy by 25% over Logistic Regression baseline, through ensemble learning and Flask deployment.
- Brain Tumor Detection and AI-Powered Chatbot** **September 2024 – December 2024**
- Engineered deep learning models (VGG16, VGG19, DenseNet) for medical image classification boosting accuracy to 95% and decreasing training time using TensorFlow, launching RESTful APIs with Flask accelerating data retrieval by 30% and enhancing system scalability, through efficient backend optimization and big data techniques.
 - Customized LLM for brain tumor chatbot increasing response relevance by 40% and improving user experience, enabling real-time medical consultation support with 90% accuracy in symptom assessment, through natural language processing and machine learning integration.
- NeuroDetectAI – Epileptic Seizure Detection** **September 2023 – November 2023**
- Built automated ML pipeline to classify seizure vs. non-seizure events from EEG data achieving 98.1% accuracy with XGBoost, outperforming SVM and Logistic Regression in precision and recall, through advanced signal processing using Discrete Wavelet Transform and ANOVA-based feature selection.
- EduPulseAI – Student Academic Success Prediction** **August 2023 - October 2023**
- Explored comprehensive dataset of 1,045 student records with 33 features achieving $R^2 = 0.75$ with Random Forest Regressor, identifying G1, G2, parental education, and study time as top predictors, through K-Means clustering with PCA revealing four student lifestyle archetypes and their academic correlations.

TECHNICAL SKILLS

Programming & Analytics:Java, C/C++, Python, R, JavaScript, SQL, TypeScript, Shell Scripting, Statistical Modeling, Machine Learning, TensorFlow, NLP, XGBoost, LightGBM, KNN/CNN

Web Development & Frameworks:Flask, React.js, Node.js, Next.js, Django, HTML, CSS, Selenium

Cloud & Infrastructure:AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, ETL Pipelines

Database & Storage:MongoDB, SQLite, SQL Server, Database Optimization

Development Tools & Methodologies:VS Code, Git, Agile/Scrum, CI/CD, DevOps, Microservices Architecture